

Time-Driven Activity-Based Costing Accurately Determines Bundle Cost for Rotator Cuff Repair



Dylan S. Koolmees, B.S., Prem N. Ramkumar, M.D., M.B.A., Cassandra Solsrud, B.S.,
Nikhil R. Yedulla, B.S., Kareem G. Elhage, B.S., Austin G. Cross, B.S., and
Eric C. Makhni, M.D., M.B.A.

Purpose: The purpose of this study was to determine the cost of the episode of care for primary rotator cuff repair (RCR) from day of surgery to 90 days postoperatively using the time-driven activity-based costing (TDABC) method. The secondary purpose of this study was to identify the main drivers of cost for both phases of care. **Methods:** This retrospective case series study used the TDABC method to determine the bundled cost of care for an RCR. First, a process map of the RCR episode of care was constructed in order to determine drivers of fixed (i.e., rent, power), direct variable (i.e., healthcare personnel), and indirect costs (i.e., marketing, building maintenance). The study was performed at a Midwestern tertiary care medical system, and patients were included in the study if they underwent an RCR from January 2018 to January 2019 with at least 90 days of postoperative follow-up. In this article, all costs were included, but we did not account for fees to provider and professional groups. **Results:** The TDABC method calculated a cost of \$10,569 for a bundled RCR, with 76% arising from the operative phase and 24% from the postoperative phase. The main driver of cost within the operative phase was the direct fixed costs, which accounted for 35% of the cost in this phase, and the largest contributor to cost within this category was the cost of implants, which accounted for 55%. In the postoperative phase of care, physical therapy visits were the greatest contributor to cost at 59%. **Conclusion:** In a bundled cost of care for RCR, the largest cost driver occurs on the day of surgery for direct fixed costs, in particular, the implant. Physical therapy represents over half of the costs of the episode of care. Better understanding the specific cost of care for RCR will facilitate optimization with appropriately designed payment models and policies that safeguard the interests of the patient, physician, and payer. **Level of Evidence:** IV, therapeutic case series.

See commentary on page 2378

Introduction

Current models have health care expenditure projected to increase at an average annual rate of 5.6% until 2025 and will encompass 19.9% of gross domestic

Henry Ford Health System, Department of Orthopedic Surgery, Detroit, Michigan, U.S.A. (D.S.K., K.S., N.R.Y., K.G.E., A.G.C., E.C.M.); and Brigham and Women's Hospital / Harvard Medical School Boston, Massachusetts, U.S.A. (P.N.R.).

Full ICMJE author disclosure forms are available for this article online, as supplementary material.

Received October 9, 2021; accepted February 6, 2022.

Address correspondence to Eric C. Makhni, M.D., M.B.A., Director of Quality and Informatics, Orthopedic Service Line Medical Director, Center for Patient-Reported Outcome Measures, Henry Ford Health System, Department of Orthopaedic Surgery, 2799 W. Grand Blvd., Detroit, MI, 48202, U.S.A. E-mail: ericmakhnimd@gmail.com.

© 2022 Published by Elsevier on behalf of the Arthroscopy Association of North America

0749-8063/211444/\$36.00

<https://doi.org/10.1016/j.arthro.2022.02.006>

product by 2025.¹ Exponential increases in the annual rates of health care cost and spending pushed health care systems to transition to value-based health care. Value-based health care prioritizes patient needs and the value delivered by clinical and surgical care, defined as health outcomes achieved per dollar spent.² In the value-based system, health care providers and administrators must devise an improved method to account for the cost of health care as a true reflection of its value. Measuring true costs in health care is difficult because of the ambiguity in customary definitions of direct and indirect care costs. Furthermore, limitations of traditional accounting methods—which can include activity-based costing (ABC), ratio of charges to cost (RCC), and relative value units (RVUs)—may compromise how accurately health care costs are measured and distributed. Using alternate accurate cost accounting methods, such as TDABC, can help providers understand the major drivers of health care

costs and ultimately result in greater value provided for patients.

Currently, there is a minimal amount of research that has investigated cost accounting in orthopedic sports medicine procedures, particularly rotator cuff repair (RCR). This is one of the most common orthopedic sports medicine procedures and will continue to be a large proportion of orthopedic sports medicine as the population continues to age.³ In order to transition to a value-based reimbursement system, costs must be thoroughly understood in order to not only define reasonable target pricing but also maximize value through optimization of cost drivers. These drivers, in the context of rotator cuff repair, are represented by the personnel labor costs for the day of surgery, physical therapy costs, and implant costs. The impact of each must be calculated. Recently, accounting tools such as time-driven activity-based cost accounting (TDABC) have helped identify and quantify these costs.

The TDABC method calculates the costs of health care resources consumed through a patient's cycle of care or total time a patient spends being treated for a certain condition. This cost accounting method focuses on two parameters: 1) cost per unit time and 2) units of time required to complete each step in a cost-dependent process.⁴ This is in contrast with traditional cost accounting methods, such as activity-based costing (ABC), which calculate costs by dividing the total expenses of said department by the quantity of each activity a department undertakes. A major limitation of the traditional ABC method is the amount of time, resources, employees, and computing power required to gather and analyze the data necessary for generating activity-specific costs.⁴ Comparatively, TDABC improves cost accounting through the use of time maps, which report the costs of an ongoing activity by displaying both the costs and time spent on a department's activities. Hence, TDABC cost models are able to measure costs without the immense amount of dedicated resources required by ABC models to generate activity-specific costs. TDABC has previously demonstrated improved accuracy in cost accounting for orthopedic procedures, such as total hip, knee, and shoulder arthroplasty.⁴⁻⁷

The purpose of this study was to determine the cost of the episode of care for primary RCR from day of surgery to 90 days postoperatively using the time driven activity-based costing (TDABC) method.⁸ The secondary purpose of this study was to identify the main drivers of cost for both phases of care. We hypothesize that procedural costs will be a substantial portion of the overall cost of RCR care, with physical therapy accounting for a majority of the postoperative cost of care.

Methods

Institutional Review Board approval was obtained from the Henry Ford Hospital (IRB no. 14091), and no external funding was used for this study. In this study, 100 patients were randomly selected to be in the cohort if they had an RCR performed from January 2018 to January 2019. Chart review was performed to gather the patient's demographic information and time required for visits. The study was performed at a Midwestern tertiary care and community care academic medical system. This institution comprises two acute tertiary care hospitals, three community care hospitals, and 17 independent clinic sites serving over three million outpatients annually.

In order to understand the cost analysis of the TDABC method, a basic foundation of accounting methods must be established. There are two main categories: 1) direct costs, which comprise both fixed and variable aspects, and 2) indirect costs.

The first category, direct costs, are those linked with a doctor or nurse delivering care to a patient. This category may further be broken down into variable costs, such as each of the differential salaries of employees rendering patient care (orthopedic surgeon, nurse, resident, and anesthesiologist). The costs in this subcategory vary with an increase or decrease in the number of patients that are aided. Conversely, the fixed direct costs are connected with doctors or nurses providing care but do not change regardless of the number of patients. This subcategory may include the cost of equipment, rent, and power necessary to ensure proper patient care.^{9,10} The last cost category, indirect costs, are those that are necessary for patient care, but are not directly related to patient care itself. Examples of indirect costs include hospital administrative salaries, marketing, and building maintenance. In this article, all costs were included, but we did not account for fees to provider and professional groups.

Creating a Process Map

The initial step of the TDABC method was to create two separate time maps for the operative and 90 days postoperative phases of a hospital-based primary RCR in order to determine the amount of time for each aspect of the episode of care that required provider interaction. The operative phase time map displays patient-related activities during the date of surgery (Fig 1), while the postoperative phase time map shows the various services used by patients following discharge from the hospital (Fig 2). The timing of the operative phase was obtained through retrospective chart review of 100 patients who underwent an RCR from January 1, 2018, to January 1, 2019. Furthermore, patients were only included in the study group if they completed their postoperative care

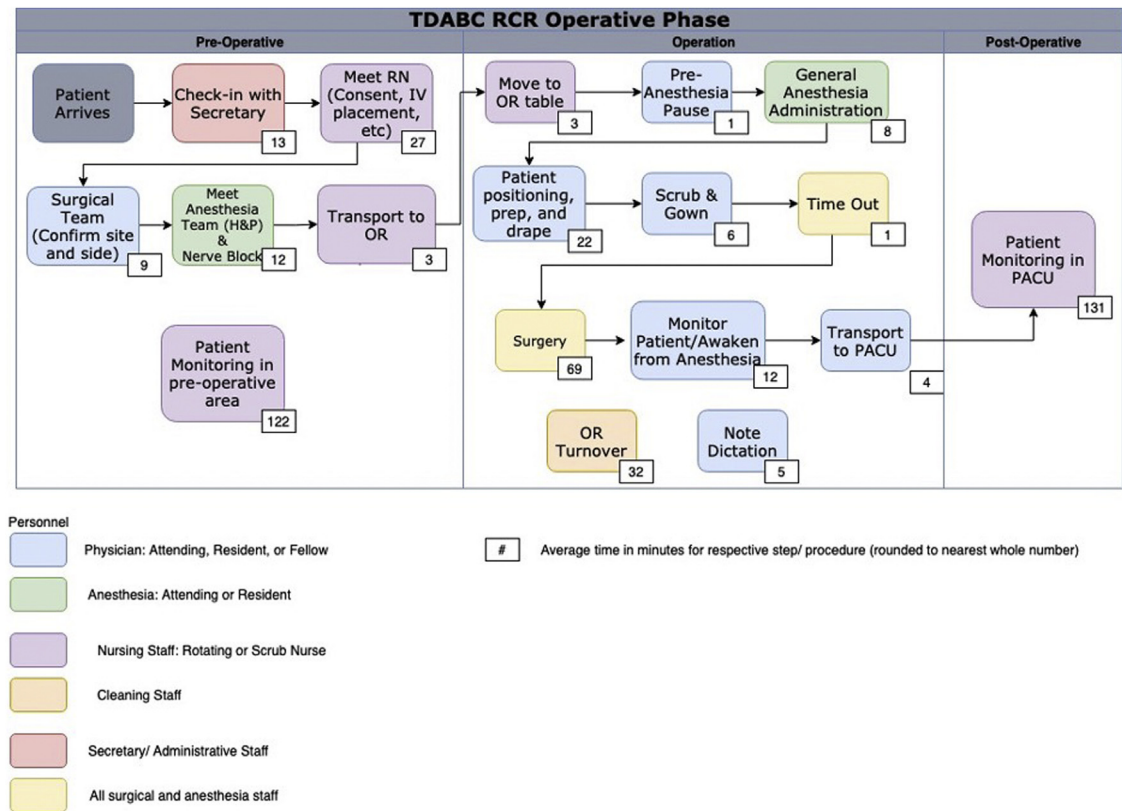


Fig 1. Time map of the operative phase for rotator cuff repair episode of care.

within the hospital system. Additionally, patients were excluded from the study if they did not receive at least 90 days of postoperative follow-up. This was to ensure that all aspects of postoperative care for an RCR were captured. The time map of the surgical episode contains all aspects of the procedure from initial check-in through discharge (Fig 1).

The time map began with patient admission into the preoperative area. The check-in process began at registration with the secretary and was followed by admission into the preoperative area, in which a nurse assisted in answering questions and placing an intra-venous catheter (IV). In our hospital, perioperative nurses typically care for two patients at a time.

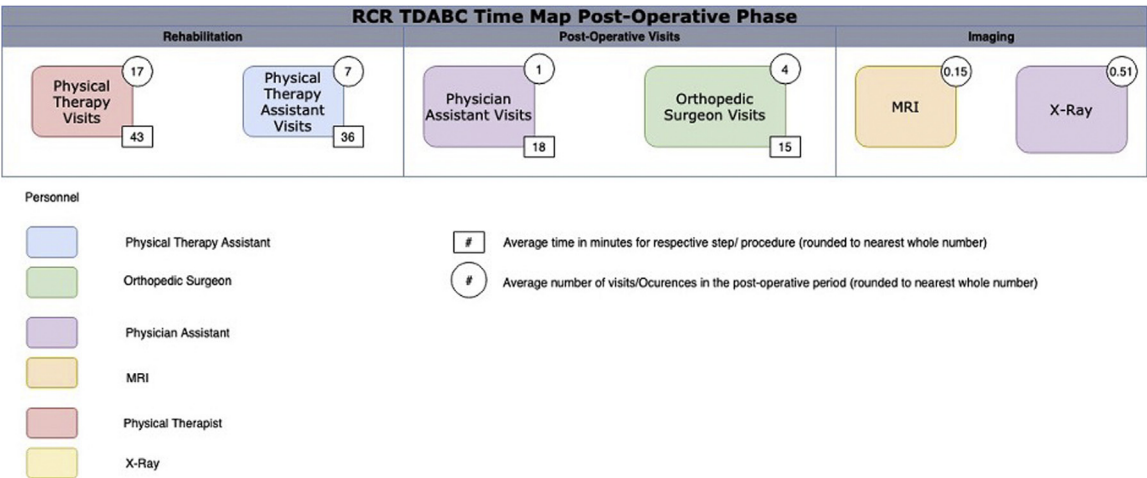


Fig 2. Time map of the postoperative phase for rotator cuff repair episode of care.

Table 1. Demographics of Patients Undergoing Rotator Cuff Repair Episode of Care

Age (years)	
Mean	59.23
Standard Deviation	10.26
Gender (<i>n</i>)	
Male	50
Female	50
BMI (kg/m ²)	
Mean	32.15
Standard Deviation	6.7
Insurance (<i>n</i>)	
Private	63
Medicare	26
Medicaid	7
Worker's Compensation	4
Race (<i>n</i>)	
White	59
Black	33
Hispanic	4
Middle Eastern	2
Native Hawaiian/Pacific Islander	1
Other	1

BMI, body mass index.

Therefore, the amount of time the nurse spent during the preoperative and postoperative periods was divided by two, in order to account for the nurse managing two patients simultaneously. The process map then includes all providers who interact with the patient, and it concludes with the time required for monitoring in the postanesthesia care unit (PACU).

The time map for the postoperative phase consisted of three major categories: 1) physical therapy visits with either a physical therapist (PT) or a physical therapist assistant (PTA); 2) postoperative medical visits with either the orthopedic surgeon or the physician assistant (PA), and 3) imaging, which mainly included radiograph and MRI. The time required for each physical therapy visit and postoperative medical visit was recorded via chart review. Then the average number of PT and PTA visits were calculated, along with the average time per visit as well. A similar process was done for the postoperative medical visits, and the average number of visits and time per visit was calculated. Finally, the average number of each type of imaging was determined, and the subsequent cost of each imaging type was generated via standard rates in the hospital's area.

Establishing Cost Drivers

The next step was to determine the cost per unit time for each of the providers that had roles in the surgical encounter, as identified in the time map. The salary of each employee from the time map was obtained from a conglomerate of salaries in the study institution's metropolitan area, using Glassdoor.¹¹ The labor rates were then condensed into a cost per minute format in

order to smoothly incorporate the time map with the costs. In addition, the practical capacity of each employee was approximated to be 80% of the theoretical maximum capacity.^{4,5} This assumption ensures that the cost per unit time is only capturing the actual time the employee spends doing his or her job.¹² In essence, this accounts for breaks and other stoppages in work that do not pertain specifically to the employee's job. Lastly, each provider was assumed to have 4 weeks of vacation per year.^{4,5} This ensures that only minutes spent actively working during the RCR are taken into account. The amount of time each provider worked during that period was then multiplied by the respective cost per unit of time in order to determine the total variable direct costs. For example, if the orthopedic surgeon spent 91 minutes of direct patient care throughout the entire process of an RCR, then the time is multiplied by his or her cost driver rate of \$6.19 per minute. In this example, the direct variable cost of the surgical episode for the orthopedic surgeon would be \$563.29, which is a separate from the facility cost. This process was repeated with each provider present throughout the process of the anterior cruciate ligament repair (registered nurse, PA, anesthesiologist, surgical resident, scrub technician, Certified Registered Nurse anesthetist, and cleaning staff). The total per-case fixed direct costs were then calculated using hospital accounting data from all RCR performed by three orthopedic surgeons from January 2018 to June 2019, with the results averaged. Each patient received one sling to use during the postoperative period, which was part of the fixed direct costs. Lastly, the cost of anchors were also obtained from the hospital's accounting department.

Lastly, the indirect costs included administrative, human resources, and other departments that were necessary for operations, but not directly tied to patient care. The TDABC method did present some challenges in terms of allocating indirect costing. The TDABC method is unable to account for every indirect cost, such as building maintenance, water, and electricity costs, as it would take an inordinate amount of time to determine each individual indirect cost. Other authors have noted this dilemma and have used a fixed ratio to determine the indirect costs for TDABC.^{5,6} This rate was determined by the average ratio of indirect to direct costs for RCR.

Results

Demographics

A total of 100 patients were included in the final analysis. The average age was 59.2 ± 10.3 (range of 33 to 78), and there were 50 males and 50 females in the study group. The average BMI was 32.2 ± 6.7 (range of 21.1 to 51.2) (Table 1). The majority of the patients had

Breakdown of Post-Operative Direct Variable Costs for Arthroscopic RCR

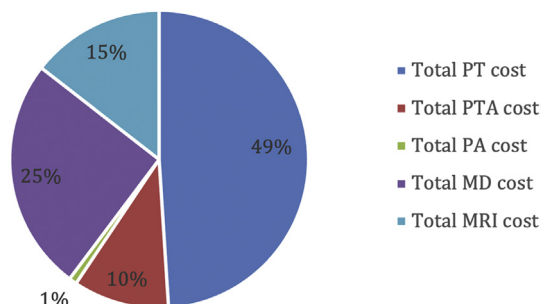


Fig 3. Breakdown of postoperative direct variable costs for rotator cuff repair episode of care.

private health insurance (63), followed by Medicare, Medicaid, and Worker's Compensation (26, 7, and 4, respectively). There were 59 Non-Hispanic/Caucasian patients, 33 Black patients, 4 Hispanic patients, 2 Middle-Eastern patients, 1 Pacific Islander patient, and 1 other.

Time Map

On the day of the RCR, time spent in the preoperative phase was composed of an average of 64 minutes between patient arrival and transport to the operating room (OR) and 122 minutes spent by nursing staff monitoring the patients in the preoperative area (Fig 1). Following the preoperative preparations, the operative phase took an average of 126 minutes total. The two longest activities of this phase were patient prepping and draping, and the surgery itself, which were 22 and 69 minutes, respectively. An additional 32 minutes were spent on OR turnover and 5 minutes on note dictation. Postoperatively, 131 minutes were spent by nursing staff monitoring patients in the PACU. Overall, nursing staff spent an average of 286 minutes with patients throughout the operative phase, the most of any provider. Lastly, there was an average of 4 arthroscopic anchors used per patient.

After their procedures, patients were most likely to visit a PT, seeing them an average of 17 times. The visits with the PT were on average 43 minutes (Fig 2). Patients also saw a PTA, seeing them an average of 7 times in the postoperative period. These visits took on average 36 minutes. During the postoperative phase, the patient saw both the orthopaedic surgeon and a physician assistant to evaluate for any postoperative complications. There were on average 4 visits with the orthopaedic surgeon and 1 with the physician assistant, and each visit was 15 minutes on average. Lastly, imaging in the postoperative period was examined, and

from the 100 patients in the sample, there were on average 0.15 MRI's and 0.51 radiographs.

Cost

In the operative phase, the largest personnel-based contribution to the overall cost of this phase came from the orthopedic surgeon. He or she accounted for 15% of the overall cost in this phase. Lastly, in the postoperative phase, the largest personnel contributor of cost was the PT, accounting for nearly half of the direct variable cost in this period (Fig 3). The next highest contributor in the postoperative period was the orthopedic surgeon at 25% of the direct variable cost (Fig 3). The cost of the surgeon was separate from the facility cost. Arthroscopic anchors contributed substantially to overall cost: at 12% of the operative costs and 9.2% of the overall cost.¹³

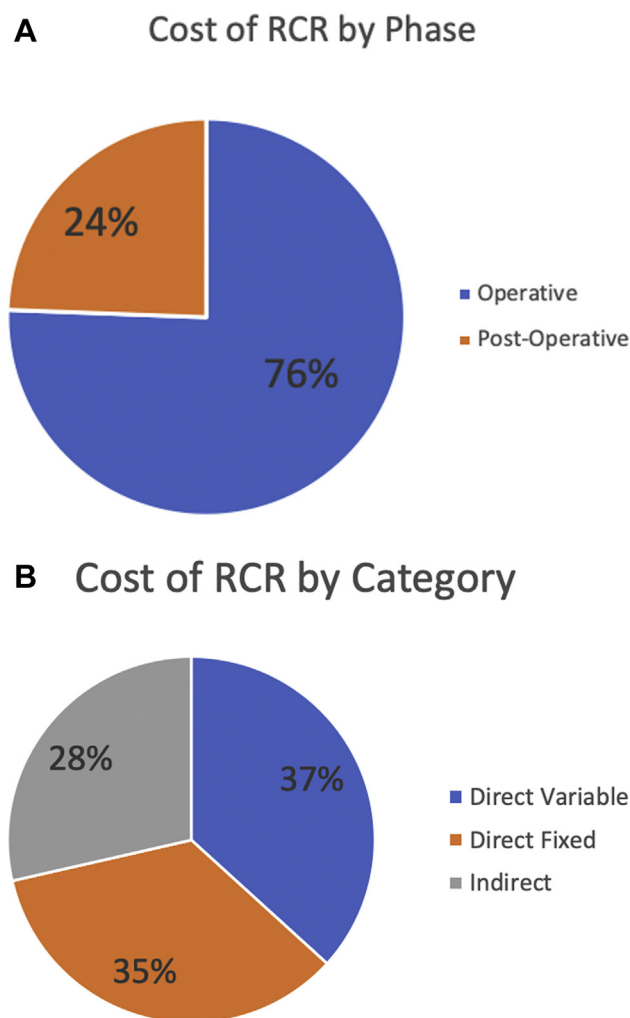


Fig 4. (A) The percent of total cost in rotator cuff repair episode of care broken down by phase. (B) The cost of a rotator cuff repair episode of care broken down by category.

The TDABC method calculated a cost of \$10,569 for RCR from day of surgery to 90 days postoperatively. When examining each phase of the surgical episode, 76% arose from day of surgery and 24% from the postoperative phase. When focusing on each category of the surgical episode, 37% of the total cost came from direct variable costs, 35% from direct fixed costs, and 28% from indirect costs (Fig 4).

Discussion

This study used the TDABC method to determine the bundled cost of care for RCR is \$10,569, and the main cost drivers were orthopedic surgeon, suture anchors, and physical therapy-related costs. Little is known about the true costs of this procedure, and our study aims to uncover and understand the costs of this procedure. Our study provides a comprehensive view of the total cost of RCR by examining the cost drivers for the procedure and the 90-day postoperative period, including physical therapy, postoperative physician visits, and any imaging needed.¹³ The largest percentage of the overall cost of the episode of care was the day of surgery, accounting for 76%. The main driver of cost within the operative phase was the direct fixed costs, which accounted for 35% of the cost in this phase, and the largest contributor to cost within this category was the cost of implants, which accounted for 55%.

Understanding costs of care is critical to the future of health care organizations, as health care transitions from fee-for-service to value-based care. This cost information represents an essential component of decision making and allows physicians, policy makers, and payers to assign value to procedures. Other cost analysis studies have looked at the cost of hip or knee arthroplasties and similarly found that the largest cost driver is from the operative phase of care (i.e., day of surgery).^{13,14} In our study, one of the largest cost drivers was the direct variable cost of the orthopedic surgeon. Previous studies have shown that the largest cost driver for RCR was the addition of subacromial decompression.¹⁵ This is in congruence with our study, as adding another aspect to the RCR procedure will lengthen the time necessary for the orthopedic surgeon to complete the procedure. As a result, our study expands on a possible reason as to why the largest cost driver in the previous study was subacromial decompression.

Additionally, the cost of the implant, a fixed direct cost, was a substantial cost driver within the procedural phase for the RCR episode of care. This is in congruence with previous studies in the field of orthopedic surgery that have shown that implant cost is a main driver of

overall cost of the operation.^{5,6,16} This presents an important avenue for surgeons and hospital administrators to reduce costs, while maintaining high value-based care. Given this knowledge of high implant cost, relative to other factors, hospitals should negotiate for improved rates to allow for efficient cost reduction.

Accurately understanding costs allows physicians and hospitals to design effective value-based health care systems and policies that benefit both the interests of the physician and of the patient. However, it is important to note that over a 10-year period from 2005 to 2014, the reimbursements for a surgeon performing a RCR has been stable, while the hospital reimbursement has substantially increased.¹⁷⁻¹⁹ Therefore, it is vital for orthopedic surgeons to understand the cost and what drives the overall cost, so that they can effectively negotiate reimbursement.

Lastly, bundled payments for orthopedic surgeries have been well documented in the field of arthroplasty.²⁰⁻²² In a study by Ryan et al., the authors examined what factors predict exceeding the cost of the bundle payment target and found multiple factors and comorbidities that could predict excess cost.²³ However, bundled payments for RCR have not yet been established.^{24,25} Our study is the first step in the process of creating a bundle payment for RCR, as a bundle payment target must first be set, in order to then determine what patient factors contribute to exceeding this limit. Therefore, more work must be done, once a consensus has been reached regarding the bundle payment target for RCR, and then future studies can elucidate what factors contribute to excess cost for patients undergoing RCR.

Limitations

Our study was not without limitations. First, the study was performed at a single institution, and results may not be generalizable on a wide scale. Secondly, there could be limitations in drawing conclusions from the overall cost of a bundled RCR, calculated using the TDABC method, due to inaccuracies in cost estimates. Another limitation was the use of Glassdoor for the salaries of the employees for the generation of the cost per unit time. However, the yearly salary of the sports medicine orthopedic surgeon was reported to be \$570,538, and a Merrit Hawkins review published data that showed nationwide orthopedic surgeons make on average \$536,000 per year.²⁶ Additionally, the average salary of an anesthesiologist in the geographical area of the study on Glassdoor is \$364,743, and the Merrit Hawkins database showed that an anesthesiologist's average annual salary was \$367,000.²⁶ Merrit Hawkins has published yearly data on the salaries of various physician specialties, and in large part can be regarded as accurate. These values are within 6% of each other

and displays that the values obtained from Glassdoor can be reliable. Lastly, because of the way that TDABC calculates the cost of RCR, it is possible that granular costs, directly and indirectly related to patient care, are missed.

Conclusion

In a bundled cost of care for RCR, the largest cost driver occurs on the day of surgery for direct fixed costs, in particular, the implant. Physical therapy represents over half of the costs of the episode of care. Better understanding the specific cost of care for RCR will facilitate optimization with appropriately designed payment models and policies that safeguard the interests of the patient, physician, and payer.

References

1. Keehan SP, Stone DA, Poisal JA, et al. National health expenditure projections, 2016-25: Price increases, aging push sector to 20 percent of economy. *Health Aff (Millwood)* 2017;36:553-563. doi:10.1377/hlthaff.2016.1627.
2. Porter ME. What is value in health care? *N Engl J Med* 2010;363:2477-2481. doi:10.1056/NEJMp1011024.
3. Colvin AC, Egorova N, Harrison AK, Moskowitz A, Flatow EL. National trends in rotator cuff repair. *J Bone Joint Surg Am* 2012;94:227-233. doi:10.2106/JBJS.J.00739.
4. Kaplan RS, Anderson SR. Time-driven activity-based costing. *Harv Bus Rev* 2004;82:131-138:150.
5. Akhavan S, Ward L, Bozic KJ. Time-driven activity-based costing more accurately reflects costs in arthroplasty surgery. *Clin Orthop Relat Res* 2016;474:8-15. doi:10.1007/s11999-015-4214-0.
6. Menendez ME, Lawler SM, Shaker J, Bassoff NW, Warner JJP, Jawa A. Time-driven activity-based costing to identify patients incurring high inpatient cost for total shoulder arthroplasty. *J Bone Joint Surg* 2018;100:2050-2056. doi:10.2106/JBJS.18.00281.
7. Palsis JA, Brehmer TS, Pellegrini VD, Drew JM, Sachs BL. The cost of joint replacement: comparing two approaches to evaluating costs of total hip and knee arthroplasty. *J Bone Joint Surg* 2018;100:326-333. doi:10.2106/JBJS.17.00161.
8. Koehler DM, Balakrishnan R, Lawler EA, Shah AS. Endoscopic versus open carpal tunnel release: A detailed analysis using time-driven activity-based costing at an academic medical center. *J Hand Surg* 2019;44:62.e1-62.e9. doi:10.1016/j.jhsa.2018.04.023.
9. Kachare SD, Liner KR, Vohra NA, Zervos EE, Hickey T, Fitzgerald TL. Assessment of health care cost for complex surgical patients: Review of cost, re-imbursement and revenue involved in pancreatic surgery at a high-volume academic medical centre. *HPB* 2015;17:311-317. doi:10.1111/hpb.12349.
10. Sharan AD, Schroeder GD, West ME, Vaccaro AR. Understanding time-driven activity-based costing. *Clin Spine Surg* 2016;29:62-65. doi:10.1097/BSD.0000000000000360.
11. "Salary Average." Glassdoor, www.Glassdoor.Com/Salaries/Estimator-Salary-SRCH_KO0.9.htm.
12. McCreary DL, White M, Vang S, Plowman B, Cunningham BP. Time-driven activity-based costing in fracture care: Is this a more accurate way to prepare for alternative payment models? *J Orthop Trauma* 2018;32:344-348. doi:10.1097/BOT.0000000000001185.
13. DiGioia AM, Greenhouse PK, Giarrusso ML, Kress JM. Determining the true cost to deliver total hip and knee arthroplasty over the full cycle of care: Preparing for bundling and reference-based pricing. *J Arthroplasty* 2016;31:1-6. doi:10.1016/j.arth.2015.07.013.
14. Koolmees D, Bernstein DN, Makhni EC. Time-driven activity-based costing provides a lower and more accurate assessment of costs in the field of orthopaedic surgery compared with traditional accounting methods. *Arthroscopy* 2021;37:1620-1627. doi:10.1016/j.arthro.2020.11.028.
15. Li L, Bokshan SL, Ready LV, Owens BD. The primary cost drivers of arthroscopic rotator cuff repair surgery: A cost-minimization analysis of 40,618 cases. *J Shoulder Elbow Surg* 2019;28:1977-1982. doi:10.1016/j.jse.2019.03.004.
16. Morris JH, Malik AT, Hatf S, Neviasser AS, Bishop JY, Cvetanovich GL. Cost of arthroscopic rotator cuff repairs is primarily driven by procedure-level factors: A single-institution analysis of an ambulatory surgery center. *Arthroscopy* 2021;37:1075-1083. doi:10.1016/j.arthro.2020.11.033.
17. Gamradt S. Editorial Commentary: Stagnant surgeon reimbursement since 2005 does not reflect the current complexity of arthroscopic rotator cuff repair in 2020. *Arthroscopy* 2020;36:2362-2363. doi:10.1016/j.arthro.2020.06.021.
18. LaPrade MD, Camp CL, Brockmeier SF, Krych AJ, Werner BC. The cost of outpatient arthroscopic rotator cuff repairs: Hospital reimbursement is on the rise while surgeon payments remain unchanged. *Arthroscopy* 2020;36:2354-2361. doi:10.1016/j.arthro.2020.03.038.
19. McIntyre LF, Beach WS. Editorial Commentary: A rigged game—Surgeon reimbursement under the resource-based relative value scale, current procedural terminology, and the Affordable Care Act. *Arthroscopy* 2020;36:2364-2366. doi:10.1016/j.arthro.2020.05.010.
20. Jiranek WA, Fehring TK, Pagnano MW, et al. Lessons learned from bundled payment programs. *Instr Course Lect* 2018;67:629-644.
21. Kee J, Mears SC, Edwards PK, Lowry Barnes C. Cost analysis and bundled care of hip and knee replacement. *J Surg Orthop Adv* 2019;28:241-249.
22. Bosco JA, Harty JH, Iorio R. Bundled payment arrangements: Keys to success. *J Am Acad Orthop Surg* 2018;26:817-822. doi:10.5435/JAAOS-D-17-00022.
23. Ryan SP, Goltz DE, Howell CB, et al. Predicting costs exceeding bundled payment targets for total joint

- arthroplasty. *J Arthroplasty* 2019;34:412-417. doi:[10.1016/j.arth.2018.11.012](https://doi.org/10.1016/j.arth.2018.11.012).
24. Dickinson RN, Kuhn JE, Bergner JL, Rizzone KH. A systematic review of cost-effective treatment of post-operative rotator cuff repairs. *J Shoulder Elbow Surg* 2017;26:915-922. doi:[10.1016/j.jse.2017.02.009](https://doi.org/10.1016/j.jse.2017.02.009).
25. Koolmees D, Ramkumar PN, Hessburg L, Guo E, Bernstein DN, Makhni EC. Time-driven activity-based costing for anterior cruciate ligament reconstruction: A comparison to traditional accounting methods. *Arthrosc Sports Med Rehabil* 2021;3:e39-e45. doi:[10.1016/j.asmr.2020.08.006](https://doi.org/10.1016/j.asmr.2020.08.006).
26. Merritt Hawkins. 2021 Review of Physician and Advanced Practitioner Recruiting Incentives, <https://www.merrithawkins.com/uploadedFiles/physician-advanced-practitioner-incentive-review-2021.pdf>. Accessed November 20, 2021.